Md Imran

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Education

2022 - 2026 B.Tech In Information Technology GPA: 7.5
Netaji Subhash Engineering College

Experience

Petition.io — Blockchain Engineer (Remote, Washington) June 2025 – Present

- Architected **Protocol-Owned Liquidity (POL)** mechanisms inspired by **Olympus DAO**, enabling sustainable treasury-backed liquidity management.
- Designed and implemented **OHM-style staking and rebasing logic**, issuing PTNR and yield-bearing stPTNR governance tokens.
- Integrated **Rocket Pool (rETH)** and **Uniswap V3 liquidity routing** to optimize protocol-controlled asset flows during staking and redemption.
- Built **quadratic funding reward distribution** pipelines using Chainlink Automation to automate yield allocation.
- Developed an **ERC-721 NFT tax receipt system** providing verifiable on-chain donation proofs and governance incentives.
- Constructed treasury accounting and **Olympus DAO-inspired metrics tracking** for protocol sustainability analysis.
- Wrote extensive **unit, integration, and invariant tests** ensuring contract security and gas efficiency.

Projects

PropToken — Solidity, Hardhat, ERC-721, ERC-20, OpenZeppelin, React, Ethers.js, MetaMask Link

- Built a **Real-World Asset (RWA) tokenization system** enabling fractional ownership using **ERC-721 property NFTs** and **ERC-20 share tokens**.
- Designed a **modular smart contract architecture** separating property minting, fractionalization, and ownership logic.
- Implemented **role-based access control** with secure, event-driven state tracking to ensure contract integrity.
- Developed comprehensive **Hardhat unit and integration test suites** covering edge cases and transactional flows.
- Delivered an **end-to-end Web3 dApp** integrating **Ethers.js wallet interaction** for real-time ownership management and token transfers.
- Applied **gas optimization strategies** and **secure smart contract development practices** to deliver **production-grade contract architecture**.

ChainVault — *Solidity, Foundry, Lit Protocol, Firebase, Node.js, Wagmi , ENS, React, Ethers.js* Link

- Built a decentralized nominee-based **inheritance system** for **secure asset recovery** via **time-locked logic**
- Integrated **Lit Protocol** for **encrypted key management** and authorized nominee access.
- Developed **modular**, **gas-optimized** smart contracts with strong **access control** and **event-driven** flow.
- Achieved **full test coverage** using **Foundry (unit, integration, fuzz, invariant tests).**
- Created a **React UI** for **vault creation, nominee management**, and **encrypted recovery workflows.**
- Built a responsive UI with **React.js** and **Redux Toolkit**, and deployed on **Vercel (frontend)** and **Render (backend)** with **CI/CD**.

Technical Skills

Programming Languages:	**Solidity, TypeScript, JavaScript, Python, C++, Java**
Blockchain & Web3:	**Foundry, Hardhat, ERC-20, ERC-721, Protocol-Owned Liquidity (POL), Staking & Rebasing Mechanisms, Uniswap V2, Uniswap V3, Rocket Pool (rETH), Chainlink Automation, Chainlink VRF, OpenZeppelin, Ethers.js, Wagmi, ENS, IPFS, Alchemy**
Frontend:	**React.js, Next.js, Tailwind CSS, Redux Toolkit, HTML, CSS**
Backend & Databases:	**Node.js, Express.js, REST APIs, Firebase, MongoDB, PostgreSQL**
DevOps & Tools:	**Git, GitHub, GitLab, Linux, AWS, Vercel, Railway, Render, CI/CD Pipelines**